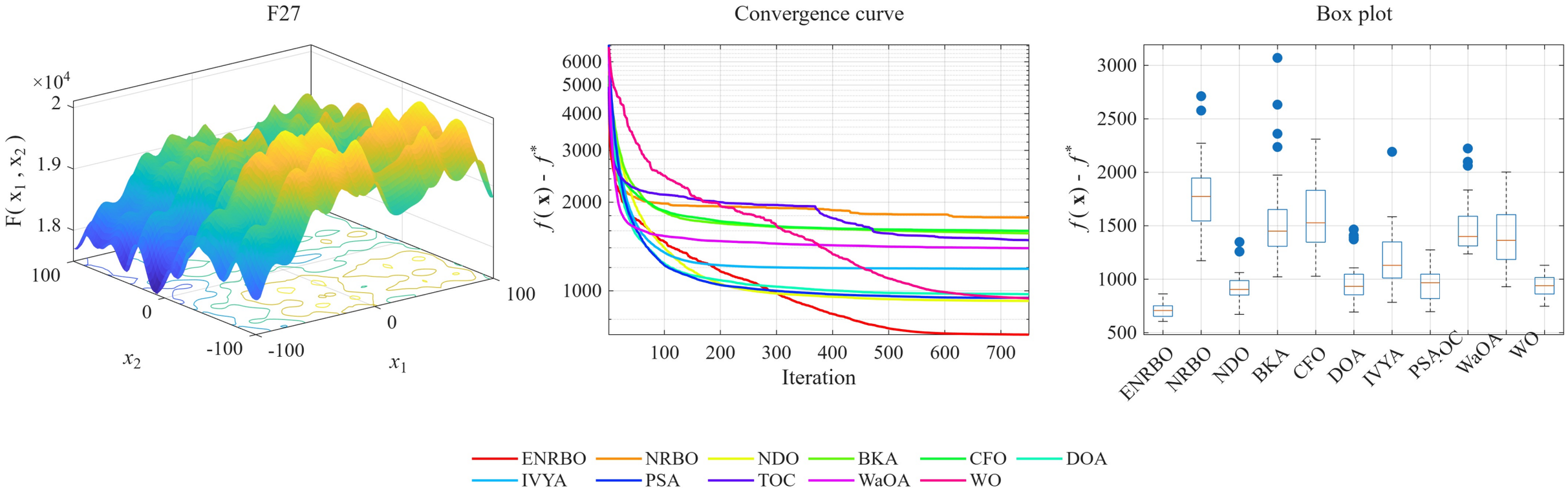

F27
×10^4
2
1.9
1.8
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
6000
5000
4000
3000
2000
1000
f (x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
3000
2500
2000
1500
1000
500
f (x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
OC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO